

# A Literature Review on the Waist Circumference Criteria for Metabolic Syndrome in Japan: Evidence and Implications for Cardiovascular Risk Stratification

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Waist circumference (WC) is widely used as a marker of central obesity in the diagnosis of metabolic syndrome (MetS) and it plays an important role in cardiovascular disease (CVD) risk screening. In Japan, the MetS diagnostic criteria established in 2005 require abdominal obesity as a mandatory component, with WC cutoff values of 85 cm for men and 90 cm for women. These criteria are also applied in the national Specific Health Checkup and Specific Health Guidance programs to identify individuals at a high risk of CVD. However, the scientific basis and clinical implications of these thresholds remain controversial.

This review summarizes the epidemiological evidence on WC cutoff values in the Japanese population and their relevance to CVD risk. A literature search of PubMed and Ichushi-Web identified studies examining WC thresholds in relation to visceral fat accumulation, clustering of cardiometabolic risk factors, and incident CVD. The evidence was organized into four themes: (1) WC cutoffs based on visceral fat area, (2) prediction of cardiometabolic risk factor clustering, (3) the association between WC and incident CVD, and (4) associations between MetS definitions and CVD risk.

Available evidence suggests that the WC thresholds corresponding to visceral fat accumulation and cardiometabolic risk clustering are generally lower than the current Japanese criterion for women. Prospective studies examining WC alone have reported inconsistent associations with incident CVD, particularly among women. Overall, the evidence suggests that CVD risk stratification frameworks requiring abdominal obesity based on WC thresholds may have limitations. Future screening strategies in Japan may benefit from greater emphasis on clustering and the overall burden of cardiometabolic risk factors.

**Key words:** Waist Circumference, Metabolic Syndrome, Cardiovascular Disease

## Introduction

Metabolic syndrome (MetS) is a chronic and progressive complex condition sharing a common pathophysiological basis, such as insulin resistance, rather than simply a clustering of risk factors, and is associated with an increased risk of cardiovascular disease (CVD) and related diseases<sup>1)</sup>. In 1988, Gerald Reaven proposed the concept of 'Syndrome X, suggesting that dyslipidemia, including hypertriglyceridemia and low high-density lipoprotein cholesterol (HDL-C), hypertension, and insulin resistance (IR), are causally linked to premature CVD<sup>2)</sup>. This concept was

revolutionary in recognizing that this cluster of abnormalities confers an increased risk of CVD, even in the absence of hyperglycemia or diabetes. Importantly, it highlighted that individuals at an elevated risk of CVD could be identified at an early stage and targeted for preventive interventions before the onset of overt hyperglycemia or diabetes, thus underscoring its substantial public health and clinical relevance.

Over the past three decades, considerable debate has continued regarding the most appropriate terminology to define this syndrome and the most appropriate way to characterize its adverse cardiovascular

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and metabolic consequences. In 1998, the World Health Organization (WHO) proposed criteria centered on IR as the underlying mechanism, which was termed the “IR syndrome”<sup>3</sup>. In 2001, the National Cholesterol Education Program–Third Adult Treatment Panel (NCEP-ATP III) committee introduced practical clinical criteria for use in primary care to identify individuals with features of what had been termed the IR syndrome<sup>4</sup>. Recognizing abdominal, particularly visceral, obesity as a key component, the panel recommended waist circumference (WC) together with four commonly measured variables: triglycerides (TG), high-density lipoprotein cholesterol (HDL-C), blood pressure, and fasting glucose. Because insulin levels were not included in this framework, the cluster of abnormalities was formally designated as “metabolic syndrome.” The presence of any three of the five components was sufficient for diagnosis. Given its practical applicability, the NCEP-ATP III definition has been the most widely utilized.

In 2005, the International Diabetes Federation (IDF) published diagnostic criteria for metabolic syndrome that required central obesity defined by WC as an essential component and emphasized the importance of establishing WC cutoff values specific to different ethnic groups<sup>5</sup>. In 2009, several major organizations jointly issued harmonized diagnostic criteria for metabolic syndrome (MetS) in a consensus statement entitled “Harmonizing the Metabolic Syndrome,” which has since served as the international standard for the MetS diagnosis<sup>6</sup>. This statement established that no single component was required, and MetS was diagnosed when any three of the five criteria were met: elevated WC, elevated triglycerides, reduced HDL-C, elevated blood pressure, and elevated fasting glucose. Uniform cutoff values were applied across all components, whereas ethnicity- and country-specific cutoff values were used for WC.

In Japan, the diagnostic criteria for metabolic syndrome were proposed in 2005<sup>7</sup>, designating abdominal obesity as a mandatory component, with WC cutoff values of 85 cm for men and 90 cm for women; this requirement has been retained despite the 2009 international harmonization of MetS definitions. Notably, the Specific Health Checkup and Specific Health Guidance program in Japan screens individuals at a high risk of CVD based on the Japanese MetS criteria, in which abdominal obesity defined by either WC or body mass index (BMI) serves as a mandatory component of the diagnostic framework<sup>8</sup>. Consequently, the WC cutoff values have substantial implications for population-level CVD prevention in Japan, as individuals who do not meet the obesity

criteria are not classified as high-risk within the screening program. Although central obesity, typically assessed by WC, is closely linked to IR and contributes to the development of cardiometabolic abnormalities, including diabetes, that increase the risk of CVD, while also being associated with the clustering of multiple cardiometabolic risk factors, this review focuses on the evidence and implications of WC as a practical tool for directly identifying individuals at high risk of CVD.

## Literature Search

We searched the PubMed and Ichushi-Web databases on February 20, 2026, to identify studies in Japanese populations addressing WC cutoffs in relation to MetS and CVD. The detailed search strategies are provided in the Supplemental Methods. To organize the evidence, we focused on two major domains—the development of WC cutoff values and their relevance to CVD risk—and grouped the eligible studies into four themes: (1) development and evaluation of the WC cutoffs based on visceral fat area (VFA), (2) predictability of WC for clustering of cardiometabolic risk factors, (3) association between WC cutoffs and CVD risk, and (4) association between the MetS criteria and CVD risk. We constructed three search strategies corresponding to these themes: search strategy 1 for Theme 1, search strategy 2 for Theme 2, and search strategy 3 for Themes 3 and 4. Studies published in either English or Japanese were included. Because Themes 3 and 4 addressed incident CVD risk, studies were eligible for these themes if they reported incident cardiovascular disease events, hazard ratios (HR) with 95% confidence intervals (CIs), and the number of incident events. Studies using CVD mortality alone as the outcome were excluded (Supplemental Methods). The records were screened in two stages: first by title and abstract and then by a full-text review.

Although this review summarized the evidence using three major MetS frameworks (Table 1)—the Japanese criteria<sup>7</sup>, IDF criteria<sup>5</sup>, and NCEP-ATP III criteria<sup>4</sup>—particular attention was given to the comparison of WC thresholds between the Japanese criteria ( $\geq 85$  cm in men and  $\geq 90$  cm in women) and the Asian-specific IDF criteria ( $\geq 90$  cm in men and  $\geq 80$  cm in women). This emphasis reflects the international recommendations that WC cutoffs should be interpreted in the context of population- and ethnicity-specific risk profiles.

**Table 1.** Metabolic syndrome criteria

|                               | Japanese criteria                | Modified Japanese criteria       | IDF criteria for Asians                             | Modified NCEP ATP III criteria 1*                   | Modified NCEP ATP III criteria 2*                   |
|-------------------------------|----------------------------------|----------------------------------|-----------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|
| Diagnostic requirement        | high WC +<br>2 and more factors  | high WC +<br>2 and more factors  | high WC +<br>2 and more factors                     | 3 and more factors                                  | 3 and more factors                                  |
| Waist circumference, cm       | 85 in men<br>90 in women         | 90 in men<br>80 in women         | 90 in men<br>80 in women                            | 85 in men<br>90 in women                            | 90 in men<br>80 in women                            |
| Triglycerides, mg/dL          | TG $\geq$ 150<br>and/or          | TG $\geq$ 150<br>and/or          | $\geq$ 150<br>or on treatment                       | $\geq$ 150<br>or on treatment                       | $\geq$ 150<br>or on treatment                       |
| HDL cholesterol, mg/dL        | HDL-C $<$ 40<br>or on treatment  | HDL-C $<$ 40<br>or on treatment  | $<$ 40 in men<br>$<$ 50 in women<br>or on treatment | $<$ 40 in men<br>$<$ 50 in women<br>or on treatment | $<$ 40 in men<br>$<$ 50 in women<br>or on treatment |
| Blood pressure, mmHg          | $\geq$ 130/85<br>or on treatment | $\geq$ 130/85<br>or on treatment | $\geq$ 130/85<br>or on treatment                    | $\geq$ 130/85<br>or on treatment                    | $\geq$ 130/85<br>or on treatment                    |
| Fasting plasma glucose, mg/dL | $\geq$ 110<br>or on treatment    | $\geq$ 110<br>or on treatment    | $\geq$ 100<br>or diagnosed type 2<br>diabetes       | $\geq$ 100<br>or on treatment                       | $\geq$ 100<br>or on treatment                       |

Abbreviations: IDF, International Diabetes Federation; NCEP ATP III, National Cholesterol Education Program Adult Treatment Panel III; WC, waist circumference; TG, triglycerides; HDL-C, high-density lipoprotein cholesterol.

\* Modified NCEP ATP III criteria 1 and 2 were defined as broad categories for NCEP ATP III-based modified definitions used in the included studies; the two categories differed only in the waist circumference cutoff. Studies using the Joint Interim Statement 2009<sup>6)</sup> were also classified under Modified NCEP ATP III criteria 2. Although a fasting glucose threshold of  $\geq$  100 mg/dL was specified for these categories, studies using the original NCEP ATP III threshold of  $\geq$  110 mg/dL were also included. Study-specific inclusion of medication use for hypertension, dyslipidemia, or diabetes was allowed.

## Evidence Review

After removing duplicates, Search Strategies 1, 2, and 3 yielded 1,252, 2,392, and 2,086 records, respectively. Ultimately, 63 unique articles met the eligibility criteria (Supplemental Methods). These articles were distributed across the four themes as follows: 14 for Theme 1, 37 for Theme 2, seven for Theme 3, and 13 for Theme 4, thus reflecting overlaps where certain articles addressed multiple thematic areas. Research activity peaked between 2006 and the mid-2010s, with a relative paucity of recent evidence.

### Theme 1. Development and Evaluation of the WC Criteria Based on the Visceral Fat Area

The Japanese WC criteria were originally developed from the concept of visceral fat obesity as a high-risk obesity phenotype by the Japan Society for the Study of Obesity. The Society's guidelines for 'obesity disease'<sup>9)</sup> established WC thresholds of 85 cm for men and 90 cm for women, corresponding to a computed tomography-measured VFA of 100 cm<sup>2</sup> based on a linear regression analysis.

Subsequent studies in various Japanese general populations have examined WC values based on VFA (Tables 2A and 2B). While the WC equivalent to a VFA of 100 cm<sup>2</sup> largely fell within the 80–90 cm range for men, it was consistently lower than 90 cm for women (Table 2A)<sup>10–18)</sup>. Furthermore, excluding

the study by Imoto *et al.*<sup>19)</sup>, which used a distinct risk factor definition including a higher blood pressure threshold and total cholesterol, optimal VFA cut-offs for clustering of cardiometabolic risk factors generally ranged from 90 to 110 cm<sup>2</sup> in men and from 60 to 70 cm<sup>2</sup> in women (Table 2B)<sup>16, 20–23)</sup>. The WC corresponding to these women-specific VFA cut-offs ranged from 77 to 86 cm<sup>20–23)</sup>. Consequently, available evidence provides limited support for a 90-cm WC cutoff in women, as both VFA-based analyses and studies of risk factor clustering have generally suggested lower corresponding WC values. Recent efforts have also explored less invasive and more cost-effective methods for direct VFA measurement, such as a bioelectrical impedance analysis<sup>24)</sup> and ultrasonography<sup>12)</sup>.

### Theme 2. Predictability of WC for the Clustering of Cardiometabolic Risk Factors

Cross-sectional studies have examined the utility of WC as a surrogate marker of visceral fat accumulation by evaluating its ability to identify clustering of cardiometabolic risk factors (Table 3). Given that visceral fat accumulation is closely linked to insulin resistance and the downstream development of hypertension, dyslipidemia, and hyperglycemia, the performance of WC in detecting risk factor clustering has been regarded as an important indicator of its clinical and public health relevance.

**Table 2A.** Waist circumference values corresponding to a visceral fat area of 100 cm<sup>2</sup>

|       | Authors, yr                                      | Participants, <i>n</i> | Age, years | VFA measurement        | Analytical method | WC corresponding to VFA 100 cm <sup>2</sup> , cm | Sensitivity | Specificity |
|-------|--------------------------------------------------|------------------------|------------|------------------------|-------------------|--------------------------------------------------|-------------|-------------|
| Men   | Ishizuka J. 2006 JP <sup>10)</sup>               | 78                     | 65.7       | CT (umbilical level)   | Linear regression | 81.7                                             | -           | -           |
|       | Kanda Y, <i>et al.</i> 2007 <sup>11)</sup>       | 258*                   | 60.4**     | CT (umbilicus to L4–5) | Linear regression | 78                                               | -           | -           |
|       | Nakanishi S, <i>et al.</i> 2007 <sup>12)</sup>   | 88                     | 66.0       | Ultrasonography        | ROC analysis      | 85                                               | 72.0%       | 83%         |
|       | Kikuchi N, <i>et al.</i> 2010 JP <sup>13)</sup>  | 868                    | ≥ 40       | CT                     | Linear regression | 85.8                                             | -           | -           |
|       | Okauchi Y, <i>et al.</i> 2010 <sup>14)</sup>     | 2,322                  | 47.9       | BIA                    | Linear regression | 84.5                                             | -           | -           |
|       | Tadokoro S, <i>et al.</i> 2010 JP <sup>15)</sup> | 190                    | 42.05      | CT (umbilical level)   | ROC analysis      | 85                                               | 89.0%       | 81.0%       |
|       | Kataoka K, <i>et al.</i> 2011 JP <sup>16)</sup>  | 625                    | 56.3       | CT (umbilical level)   | ROC analysis      | 88.6                                             | 69.0%       | 79.6%       |
|       | Inomoto A, <i>et al.</i> 2014 JP <sup>17)</sup>  | 124                    | 49.7       | BIA                    | ROC analysis      | 87.9                                             | 80.8%       | 79.6%       |
|       | Kaya M, <i>et al.</i> 2015 <sup>18)</sup>        | 55                     | 42.0       | BIA                    | Linear regression | 84.4                                             | -           | -           |
| Women | Ishizuka J. 2006 JP <sup>10)</sup>               | 111                    | 69.6       | CT (umbilical level)   | Linear regression | 85.6                                             | -           | -           |
|       | Kanda Y, <i>et al.</i> 2007 <sup>11)</sup>       | 161*                   | 60.4**     | CT (umbilicus to L4–5) | Linear regression | 82                                               | -           | -           |
|       | Nakanishi S, <i>et al.</i> 2007 <sup>12)</sup>   | 64                     | 65.0       | Ultrasonography        | ROC analysis      | 84                                               | 75%         | 70%         |
|       | Kikuchi N, <i>et al.</i> 2010 JP <sup>13)</sup>  | 129                    | ≥ 40       | CT                     | Linear regression | 86.7                                             | -           | -           |
|       | Okauchi Y, <i>et al.</i> 2010 <sup>14)</sup>     | 548                    | 46.8       | BIA                    | Linear regression | 89.4                                             | -           | -           |
|       | Tadokoro S, <i>et al.</i> 2010 JP <sup>15)</sup> | 121                    | 42.01      | CT (umbilical level)   | ROC analysis      | 83                                               | 93.0%       | 76.0%       |

\* Participants: individuals with type 2 diabetes only    \*\* Age: mean age of men and women combined.

Abbreviations: JP, articles published in Japanese; VFA, visceral fat area; CT, computed tomography; BIA, bioelectrical impedance analysis; ROC, receiver operating characteristic.

**Table 2B.** Optimal visceral fat area cutoffs for cardiometabolic risk clustering and the corresponding waist circumference values

|       | Authors, yr                                      | Participants, <i>n</i> | Age, years | VFA measurement      | Risk-clustering criteria                                                                                                                                                                                                      | Optimal VFA cutoff, cm <sup>2</sup> | Sensitivity | Specificity | Analytical method | Corresponding WC, cm |
|-------|--------------------------------------------------|------------------------|------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------------|-------------|-------------------|----------------------|
| Men   | Miyawaki T, <i>et al.</i> 2005 <sup>20)</sup>    | 164                    | 49.1       | CT (umbilical level) | Japanese criteria (excluding WC): risk factors ≥ 2                                                                                                                                                                            | 100                                 | 67%         | 60%         | Linear regression | 86                   |
|       | Eguchi M, <i>et al.</i> 2007 <sup>21)</sup>      | 235                    | 63         | CT (umbilical level) | Japanese criteria (excluding WC): risk factors ≥ 2                                                                                                                                                                            | 92                                  | 61.2%       | 67.3%       | -                 | 83.2                 |
|       | Fujikawa R, <i>et al.</i> 2009 JP <sup>22)</sup> | 1,101                  | 46.1       | CT                   | Japanese criteria (excluding WC): risk factors ≥ 2                                                                                                                                                                            | 94                                  | 69%         | 69%         | Linear regression | 86                   |
|       | Kashihara H, <i>et al.</i> 2009 <sup>23)</sup>   | 5,080                  | 49.2       | CT (umbilical level) | Japanese criteria (excluding WC): risk factors ≥ 2                                                                                                                                                                            | 103                                 | 68.7%       | 61.8%       | Linear regression | 89.1                 |
|       | Imoto T, <i>et al.</i> 2010 JP <sup>19)</sup>    | 8,470                  | 47.3       | CT (umbilical level) | Two or more of the following factors:<br>1) SBP ≥ 140 mmHg and/or DBP ≥ 90 mmHg<br>2) total cholesterol ≥ 220 mg/dL and/or HDL-cholesterol < 40 mg/dL and/or triglyceride ≥ 150 mg/dL<br>3) fasting blood glucose ≥ 110 mg/dL | 76.3                                | 68.9%       | 66.2%       | Linear regression | 85.2                 |
|       | Kataoka K, <i>et al.</i> 2011 JP <sup>16)</sup>  | 625                    | 56.3       | CT (umbilical level) | Japanese criteria (excluding WC): risk factors ≥ 2                                                                                                                                                                            | 111.7                               | 73.8%       | 61.4%       | ROC analysis      | 89                   |
| Women | Miyawaki T, <i>et al.</i> 2005 <sup>20)</sup>    | 53                     | 49.6       | CT (umbilical level) | Japanese criteria (excluding WC): risk factors ≥ 2                                                                                                                                                                            | 65                                  | 73%         | 70%         | Linear regression | 77                   |
|       | Eguchi M, <i>et al.</i> 2007 <sup>21)</sup>      | 185                    | 61         | CT (umbilical level) | Japanese criteria (excluding WC): risk factors ≥ 2                                                                                                                                                                            | 62.8                                | 50.7%       | 60.8%       | -                 | 77.9                 |
|       | Fujikawa R, <i>et al.</i> 2009 JP <sup>22)</sup> | 516                    | 45.3       | CT                   | Japanese criteria (excluding WC): risk factors ≥ 2                                                                                                                                                                            | 60                                  | 77%         | 77%         | Linear regression | 82.4                 |
|       | Kashihara H, <i>et al.</i> 2009 <sup>23)</sup>   | 1,656                  | 51.4       | CT (umbilical level) | Japanese criteria (excluding WC): risk factors ≥ 2                                                                                                                                                                            | 69                                  | 80.8%       | 70.0%       | Linear regression | 86.3                 |

(Cont. Table 2B)

|       | Authors, yr                                   | Participants, <i>n</i> | Age, years | VFA measurement      | Risk-clustering criteria                                                                                                                                                                                                                                 | Optimal VFA cutoff, cm <sup>2</sup> | Sensitivity | Specificity | Analytical method | Corresponding WC, cm |
|-------|-----------------------------------------------|------------------------|------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------------|-------------|-------------------|----------------------|
| Women | Imoto T, <i>et al.</i> 2010 JP <sup>19)</sup> | 1,626                  | 47.5       | CT (umbilical level) | Two or more of the following factors:<br>1) SBP $\geq$ 140 mmHg and/or DBP $\geq$ 90 mmHg<br>2) total cholesterol $\geq$ 220 mg/dL and/or HDL-cholesterol $<$ 40 mg/dL and/or triglyceride $\geq$ 150 mg/dL<br>3) fasting blood glucose $\geq$ 110 mg/dL | 47.3                                | 69.8%       | 70.2%       | Linear regression | 80.1                 |

Abbreviations: VFA, visceral fat area; WC, waist circumference; JP, articles published in Japanese; CT, computed tomography; ROC, receiver operating characteristic; HDL, high-density lipoprotein.

Across the studies summarized in **Table 3**, the optimal WC cutoffs for identifying the presence of  $\geq 2$  cardiometabolic risk factors were distributed between 80 and 90 cm in men (median, 85 cm) and between 73 and 85 cm in women (median, 80 cm)<sup>14, 16, 19, 25-55</sup>. In women, the sensitivity of the 90-cm threshold was consistently low, generally ranging from 30% to 50%<sup>29, 33, 35, 42, 44, 47, 56-58</sup>. Several studies also attempted to refine these thresholds by incorporating anthropometric or metabolic parameters such as BMI<sup>59, 60</sup>, age<sup>48, 52, 61</sup>, and HOMA-IR<sup>62, 63</sup>. Concurrently, these analyses underscored the presence of metabolically obese, normal-weight individuals, indicating that metabolically high-risk states may be present even among those with a normal WC.

In summary, these findings suggest that WC is useful as a cross-sectional screening marker for clustered cardiometabolic risk. However, the reported cutoff values showed considerable variability across studies, particularly in women. Additionally, the cross-sectional nature of these analyses precludes direct inference regarding the WC threshold most appropriate for predicting future CVD events.

### Theme 3. Association between WC Cutoffs and CVD Risk

Prospective cohort studies in the Japanese general population and in patients with type 2 diabetes have examined the association between WC and CVD risk (**Tables 4A and 4B**). In men, associations with CVD risk were observed for a WC cutoff of 90 cm in several studies, whereas such associations were less often observed for a cutoff of 85 cm<sup>64-70</sup>. For example, this pattern was seen in the Hisayama Study<sup>65</sup>: a WC cutoff of 90 cm was associated with CVD risk (HR, 1.81 [95% CI, 1.19–2.74]), whereas no such association was observed for a cutoff of 85 cm (HR, 1.22 [95% CI, 0.86–1.73]). In addition, point estimates of the HR tended to be slightly higher for the 90-cm cutoff than for the 85-cm cutoff. In

contrast, findings in women were less consistent when comparing 80 cm and 90 cm thresholds<sup>64-68, 70</sup>. In the Suita Study<sup>67</sup>, among women aged  $\geq 65$  years, a WC cutoff of 90 cm was associated with incident ischemic stroke (HR, 2.54 [95% CI, 1.01–6.40]), whereas no such association was observed in younger women. Overall, however, associations between WC and CVD risk in women varied across studies.

In addition to studies comparing predefined WC thresholds, some prospective studies have attempted to derive optimal WC cutoffs for predicting incident CVD. For example, the Hisayama Study<sup>65</sup> reported optimal thresholds of 80.2 cm for men and 81.5 cm for women for predicting future CVD. Similarly, a recent study using a large administrative claims database<sup>68, 71</sup> reported that the optimal WC cutoff values for predicting coronary artery disease and stroke were 83.7 cm (or approximately 83 cm) for men and 77 cm for women. These estimates should be interpreted cautiously because they may be influenced by study-specific factors, such as follow-up duration, age distribution, and event rates.

In conclusion, these findings suggest that the association between WC and incident CVD is relatively more consistent in men than in women. In women, both comparisons of predefined thresholds and attempts to derive optimal cutoffs yielded less consistent findings across studies. Thus, the role of central obesity, as assessed by WC alone, in CVD risk stratification remains uncertain in the Japanese population, particularly among women.

### Theme 4. Association between the MetS Criteria and CVD Risk

Prospective studies have examined the association between MetS definitions and incident CVD using several diagnostic frameworks, including the Japanese, modified Japanese, IDF, and modified NCEP-ATP III criteria (**Tables 1, 5A, 5B, and 5C**). Overall, comparisons across these definitions have yielded

**Table 3.** Predictability of waist circumference for clustering of cardiometabolic risk factors

| sex | Authors, yr                                        | Participants, n | Age, years | WC threshold type | Criteria for cardiometabolic risk factor clustering                                                                                                                                                                                                                                                                         | WC cutoff, cm | Sensitivity        | Specificity |
|-----|----------------------------------------------------|-----------------|------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|-------------|
| Men | Ito H, <i>et al.</i> 2003 <sup>25)</sup>           | 768             | 44.6       | Optimized cutoff  | Two or more of the following factors:<br>1) SBP $\geq$ 140 mmHg and/or DBP $\geq$ 90 mmHg, or treatment<br>2) total cholesterol $\geq$ 240 mg/dL and/or HDL-cholesterol $<$ 40 mg/dL and/or triglyceride $\geq$ 150 mg/dL, or treatment<br>3) fasting blood glucose $\geq$ 126 mg/dL and/or HbA1c $\geq$ 6.5%, or treatment | 85            | 69.0%              | 69.0%       |
|     | Chiba Y, <i>et al.</i> 2006 <sup>26)</sup>         | 353             | 62.8       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 86            | 56.0%              | 58.0%       |
|     | Hsieh SD, <i>et al.</i> 2006 <sup>56)</sup>        | 6,141           | 49.5       | Japanese criteria | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 85            | 74.3%              | 52.7%       |
|     | Hara K, <i>et al.</i> 2006 <sup>27)</sup>          | 408             | 52.3       | Optimized cutoff  | NCEP-ATP III criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                 | 85            | 71%                | 69.8%       |
|     | Ohbayashi H, <i>et al.</i> 2006 <sup>28)</sup>     | 1,408           | 50         | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 85.0          | 75.0%              | 56.4%       |
|     | Ohbayashi H, <i>et al.</i> 2006 <sup>28)</sup>     | 1,408           | 50         | Optimized cutoff  | IDF criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                          | 85.0          | 71.7%              | 59.1%       |
|     | Lee JS, <i>et al.</i> 2007 <sup>29)</sup>          | 1,146           | 53.2       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 80            | 92.8%              | 32.1%       |
|     | Lee JS, <i>et al.</i> 2007 <sup>29)</sup>          | 1,146           | 53.2       | Optimized cutoff  | IDF criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                          | 81            | 85.8%              | 39.6%       |
|     | Lee JS, <i>et al.</i> 2007 <sup>29)</sup>          | 1,146           | 53.2       | Japanese criteria | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 85            | 64.6%              | 57.4%       |
|     | Miyatake N, <i>et al.</i> 2007 <sup>30)</sup>      | 1,252           | 46.7       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 85            | 66.3%              | 62.1%       |
|     | Nishimura R, <i>et al.</i> 2007 <sup>31)</sup>     | 955             | 51.7       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 84.9          | 63.3%              | 64.1%       |
|     | Shimajiri T, <i>et al.</i> 2007, JP <sup>32)</sup> | 1,921           | 46.8       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 86            | Euclidean distance |             |
|     | Kato M, <i>et al.</i> 2008 <sup>57)</sup>          | 315             | 62         | Japanese criteria | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 85            | 62.0%              | 59.0%       |
|     | Matoba Y, <i>et al.</i> 2008 <sup>33)</sup>        | 1,658           | 48.8       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 87            | 70.0%              | 57.0%       |
|     | Matoba Y, <i>et al.</i> 2008 <sup>33)</sup>        | 1,658           | 48.8       | Optimized cutoff  | IDF criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                          | 87            | 68.0%              | 62.0%       |
|     | Matoba Y, <i>et al.</i> 2008 <sup>33)</sup>        | 1,658           | 48.8       | Japanese criteria | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 85            | 77.0%              | 46.0%       |
|     | Narisawa S, <i>et al.</i> 2008 <sup>34)</sup>      | 7,762           | 50.7       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 87            | 66.0%              | 62.0%       |
|     | Oka R, <i>et al.</i> 2008 <sup>35)</sup>           | 1,061           | 50.9*      | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 89.8          | 44.6%              | 76.1%       |
|     | Oka R, <i>et al.</i> 2008 <sup>35)</sup>           | 1,061           | 50.9*      | Japanese criteria | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 85            | 69.9%              | 45.1%       |
|     | Sato A, <i>et al.</i> 2008 <sup>36)</sup>          | 118             | 63.6       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 87            | 62.5%              | 75.0%       |
|     | Sato A, <i>et al.</i> 2008 <sup>36)</sup>          | 118             | 63.6       | Optimized cutoff  | NCEP-ATP III criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                 | 86.5          | 69.0%              | 74.2%       |
|     | Shimajiri T, <i>et al.</i> 2008 <sup>37)</sup>     | 3,148           | 47         | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 86            | 71.0%              | 63.0%       |
|     | Kawai K, <i>et al.</i> 2009, JP <sup>38)</sup>     | 4,506           | 20-89y     | Optimized cutoff  | Two or more of the following factors:<br>1) fasting blood glucose $\geq$ 110 mg/dL or treatment<br>2) SBP $\geq$ 130 mmHg and/or DBP $\geq$ 85 mmHg, or treatment<br>3) HDL-cholesterol $<$ 40 mg/dL and/or LDL-cholesterol $\geq$ 140 mg/dL and/or triglyceride $\geq$ 150 mg/dL, or treatment                             | 84            | Youden index       |             |
|     | Takahashi M, <i>et al.</i> 2009 <sup>39)</sup>     | 3,915           | 50.8       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 85            | 68.0%              | 70.0%       |
|     | Yoshida D, <i>et al.</i> 2009 <sup>40)</sup>       | 3,758           | 62.3       | Optimized cutoff  | Two or more of the following factors:<br>1) (SBP $\geq$ 130 mmHg or DBP $\geq$ 85 mmHg) or medical treatment<br>2) non-HDL-cholesterol $>$ 170 mg/dL or medical treatment<br>3) HDL-cholesterol $<$ 40 mg/dL<br>4) HbA1c $>$ 5.8% or medical treatment                                                                      | 85            | 68.0%              | 49.7%       |
|     | Imoto T, <i>et al.</i> 2010, JP <sup>19)</sup>     | 8,470           | 47.3       | Optimized cutoff  | Two or more of the following factors:<br>1) SBP $\geq$ 140 mmHg and/or DBP $\geq$ 90 mmHg<br>2) total cholesterol $\geq$ 220 mg/dL and/or HDL-cholesterol $<$ 40 mg/dL and/or triglyceride $\geq$ 150 mg/dL<br>3) fasting blood glucose $\geq$ 110 mg/dL                                                                    | 84.2          | 65.7%              | 61.0%       |

(Cont. Table 3)

| sex | Authors, yr                                     | Participants, <i>n</i> | Age, years | WC threshold type | Criteria for cardiometabolic risk factor clustering                                                                                                                                                                                                                                                                                                            | WC cutoff, cm | Sensitivity                          | Specificity |
|-----|-------------------------------------------------|------------------------|------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------|-------------|
| Men | Oda E, <i>et al.</i> 2010 <sup>41)</sup>        | 2,444                  | 51.9       | Optimized cutoff  | NCEP-ATP III criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                    | 85            | 66.0%                                | 63.0%       |
|     | Ogawa D, <i>et al.</i> 2010 <sup>42)</sup>      | 3,811                  | 49.9       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 84            | 66.3%                                | 56%         |
|     | Ogawa D, <i>et al.</i> 2010 <sup>42)</sup>      | 3,811                  | 49.9       | Japanese criteria | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 85            | 64.2%                                | 60.2%       |
|     | Okauchi Y, <i>et al.</i> 2010 <sup>14)</sup>    | 2,322                  | 47.9       | Optimized cutoff  | Two or more of the following factors:<br>1) (SBP $\geq 130$ mmHg and/or DBP $\geq 85$ mmHg) or medical treatment<br>2) (fasting triglyceride $\geq 1.69$ mmol/L or postprandial $\geq 2.27$ mmol/L), and/or HDL-C $< 1.04$ mmol/L, or medical treatment<br>3) fasting serum glucose $\geq 6.10$ mmol/L or postprandial $\geq 7.77$ mmol/L or medical treatment | 84.6          | Maximal sensitivity plus specificity |             |
|     | Satoh H, <i>et al.</i> 2010 <sup>43)</sup>      | 4,599                  | 53         | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 86            | 64.0%                                | 62.0%       |
|     | Satoh H, <i>et al.</i> 2010 <sup>44)</sup>      | 4,344                  | 52         | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 86            | 62.0%                                | 64.0%       |
|     | Yamamoto Y, <i>et al.</i> 2010 <sup>45)</sup>   | 7,136                  | 51.2       | Optimized cutoff  | Three or more of the following factors:<br>1) SBP $\geq 130$ mmHg and/or DBP $\geq 85$ mmHg, or treatment<br>2) FPG $\geq 100$ mg/dL or HbA1c $\geq 5.2\%$ , or treatment<br>3) TG $\geq 150$ mg/dL and/or HDL-C $< 40$ mg/dL, or treatment                                                                                                                    | 83.1          | 76.8%                                | 70.7%       |
|     | Kataoka K, <i>et al.</i> 2011 JP <sup>16)</sup> | 625                    | 56.3       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 89.5          | 57.7%                                | 64.8%       |
|     | Kawada T, <i>et al.</i> 2011 <sup>46)</sup>     | 4,572                  | 42.5       | Optimized cutoff  | Two or more of the following factors:<br>1) (TG $> 150$ mg/dL and/or HDL-C $< 40$ mg/dL) or medical treatment<br>2) FPG $> 110$ mg/dL or medical treatment<br>3) (SBP $\geq 130$ mmHg and/or DBP $\geq 85$ mmHg) or medical treatment                                                                                                                          | 83.7          | 73.8%                                | 59.8%       |
|     | Nakamura K, <i>et al.</i> 2011 <sup>47)</sup>   | 330                    | 52.5       | Optimized cutoff  | Two or more of the following factors:<br>1) SBP $\geq 130$ mmHg and/or DBP $\geq 85$ mmHg<br>2) HDL-cholesterol $< 40$ mg/dL and/or TG $\geq 150$ mg/dL<br>3) HbA1c (%) $\geq 5.5$                                                                                                                                                                             | 88            | 60.0%                                | 70.0%       |
|     | Nakamura K, <i>et al.</i> 2011 <sup>47)</sup>   | 330                    | 52.5       | Japanese criteria | Two or more of the following factors:<br>1) SBP $\geq 130$ mmHg and/or DBP $\geq 85$ mmHg<br>2) HDL-cholesterol $< 40$ mg/dL and/or TG $\geq 150$ mg/dL<br>3) HbA1c (%) $\geq 5.5$                                                                                                                                                                             | 85            | 72.0%                                | 53.0%       |
|     | Matsushita Y, <i>et al.</i> 2012 <sup>48)</sup> | 9,867                  | 51.9       | Optimized cutoff  | NCEP-ATP III criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                    | 83.7          | 77.1%                                | 49.8%       |
|     | Takahara M, <i>et al.</i> 2012 <sup>49)</sup>   | 2,849                  | 48.4       | Optimized cutoff  | Two or more of the following factors:<br>1) (SBP $\geq 130$ mmHg or DBP $\geq 85$ mmHg) or medical treatment<br>2) HDL-C $< 1.04$ mmol/L, (fasting triglyceride $\geq 1.69$ mmol/L or postprandial $\geq 2.27$ mmol/L), or medical treatment<br>3) fasting serum glucose $\geq 6.1$ mmol/L or postprandial $\geq 7.8$ mmol/L, or medical treatment             | 85            | Euclidean distance                   |             |
|     | Unno M, <i>et al.</i> 2012 <sup>50)</sup>       | 546                    | 53.5       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 88.5          | 73.0%                                | 60.8%       |
|     | Hosono A, <i>et al.</i> 2013 <sup>51)</sup>     | 1,195                  | 63         | Optimized cutoff  | NCEP-ATP III criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                    | 81            | 63%                                  | 58.4%       |
|     | Okazawa T, <i>et al.</i> 2013 <sup>52)</sup>    | 7,611                  | 47.5       | Optimized cutoff  | IDF criteria (excluding WC): risk factors $\geq 2$ in non-medicated participants                                                                                                                                                                                                                                                                               | 83.1          | 68.4%                                | 60.3%       |
|     | Tsukiyama H, <i>et al.</i> 2016 <sup>53)</sup>  | 239                    | 54         | Optimized cutoff  | Joint Interim Statement 2009 <sup>6)</sup> (excluding WC): risk factors $\geq 1$                                                                                                                                                                                                                                                                               | 82.1          | Maximal sensitivity plus specificity |             |
|     | Takase H, <i>et al.</i> 2020 <sup>54)</sup>     | 8,531                  | 42.5       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 84.2          | 72.9%                                | 64.7%       |

(Cont. Table 3)

| sex   | Authors, yr                                        | Participants, n | Age, years | WC threshold type | Criteria for cardiometabolic risk factor clustering                                                                                                                                                                                                                                                                         | WC cutoff, cm | Sensitivity        | Specificity |
|-------|----------------------------------------------------|-----------------|------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|-------------|
| Women | Ito H, <i>et al.</i> 2003 <sup>25)</sup>           | 1,960           | 45.1       | Optimized cutoff  | Two or more of the following factors:<br>1) SBP $\geq$ 140 mmHg and/or DBP $\geq$ 90 mmHg, or treatment<br>2) total cholesterol $\geq$ 240 mg/dL and/or HDL-cholesterol $<$ 40 mg/dL and/or triglyceride $\geq$ 150 mg/dL, or treatment<br>3) fasting blood glucose $\geq$ 126 mg/dL and/or HbA1c $\geq$ 6.5%, or treatment | 73            | 70.0%              | 70.0%       |
|       | Chiba Y, <i>et al.</i> 2006 <sup>26)</sup>         | 457             | 57.8       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 84            | 62.0%              | 62.0%       |
|       | Hsieh SD, <i>et al.</i> 2006 <sup>56)</sup>        | 2,137           | 51.9       | Japanese criteria | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 90            | 17.2%              | 95.9%       |
|       | Hara K, <i>et al.</i> 2006 <sup>27)</sup>          | 284             | 53.5       | Optimized cutoff  | NCEP-ATP III criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                 | 78            | 60.0%              | 77.1%       |
|       | Ohbayashi H, <i>et al.</i> 2006 <sup>28)</sup>     | 932             | 49.3       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 79.5          | 78.7%              | 70.9%       |
|       | Ohbayashi H, <i>et al.</i> 2006 <sup>28)</sup>     | 932             | 49.3       | Optimized cutoff  | IDF criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                          | 78.0          | 73.9%              | 68.3%       |
|       | Lee JS, <i>et al.</i> 2007 <sup>29)</sup>          | 1,330           | 50.9       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 82            | 83.2%              | 62.1%       |
|       | Lee JS, <i>et al.</i> 2007 <sup>29)</sup>          | 1,330           | 50.9       | Optimized cutoff  | IDF criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                          | 82            | 79.6%              | 65.1%       |
|       | Lee JS, <i>et al.</i> 2007 <sup>29)</sup>          | 1,330           | 50.9       | Japanese criteria | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 90            | 52.2%              | 87.8%       |
|       | Miyatake N, <i>et al.</i> 2007 <sup>30)</sup>      | 1,933           | 48.5       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 75            | 68.2%              | 68.7%       |
|       | Nishimura R, <i>et al.</i> 2007 <sup>31)</sup>     | 1,158           | 51.7       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 81.1          | 67.0%              | 67.7%       |
|       | Shimajiri T, <i>et al.</i> 2007, JP <sup>32)</sup> | 1,411           | 43.6       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 81            | Euclidean distance |             |
|       | Kato M, <i>et al.</i> 2008 <sup>57)</sup>          | 314             | 62         | Japanese criteria | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 90            | 34.0%              | 89.0%       |
|       | Matoba Y, <i>et al.</i> 2008 <sup>33)</sup>        | 1,116           | 46.8       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 80            | 84.0%              | 68.0%       |
|       | Matoba Y, <i>et al.</i> 2008 <sup>33)</sup>        | 1,116           | 46.8       | Optimized cutoff  | IDF criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                          | 80            | 74.0%              | 70.0%       |
|       | Matoba Y, <i>et al.</i> 2008 <sup>33)</sup>        | 1,116           | 46.8       | Japanese criteria | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 90            | 38.0%              | 92.0%       |
|       | Narisawa S, <i>et al.</i> 2008 <sup>34)</sup>      | 4,963           | 49.7       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 83            | 73.0%              | 70.0%       |
|       | Oka R, <i>et al.</i> 2008 <sup>35)</sup>           | 809             | 50.9*      | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 82.3          | 71.7%              | 58.5%       |
|       | Oka R, <i>et al.</i> 2008 <sup>35)</sup>           | 809             | 50.9*      | Japanese criteria | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 90            | 32.1%              | 86.2%       |
|       | Sato A, <i>et al.</i> 2008 <sup>36)</sup>          | 277             | 63.4       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 80            | 49.0%              | 66.8%       |
|       | Sato A, <i>et al.</i> 2008 <sup>36)</sup>          | 277             | 63.4       | Optimized cutoff  | NCEP-ATP III criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                 | 80            | 61.1%              | 70.0%       |
|       | Shimajiri T, <i>et al.</i> 2008 <sup>37)</sup>     | 2,423           | 44         | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 81            | 74.0%              | 76.0%       |
|       | Kawai K, <i>et al.</i> 2009, JP <sup>38)</sup>     | 2,300           | 21-78y     | Optimized cutoff  | Two or more of the following factors:<br>1) fasting blood glucose $\geq$ 110 mg/dL or treatment<br>2) SBP $\geq$ 130 mmHg and/or DBP $\geq$ 85 mmHg, or treatment<br>3) HDL-cholesterol $<$ 40 mg/dL and/or LDL-cholesterol $\geq$ 140 mg/dL and/or triglyceride $\geq$ 150 mg/dL, or treatment                             | 80            | Youden index       |             |
|       | Narisawa S, <i>et al.</i> 2009, JP <sup>55)</sup>  | 4,953           | 49.7       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 83            | 73.0%              | 70.0%       |
|       | Takahashi M, <i>et al.</i> 2009 <sup>39)</sup>     | 2,032           | 50.7       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq$ 2                                                                                                                                                                                                                                                                     | 80            | 75.0%              | 69.0%       |
|       | Yoshida D, <i>et al.</i> 2009 <sup>40)</sup>       | 4,517           | 62.5       | Optimized cutoff  | Two or more of the following factors:<br>1) (SBP $\geq$ 130 mmHg or DBP $\geq$ 85 mmHg) or medical treatment<br>2) non-HDL-cholesterol $>$ 170 mg/dL or medical treatment<br>3) HDL-cholesterol $<$ 40 mg/dL<br>4) HbA1c $>$ 5.8% or medical treatment                                                                      | 85            | 58.8%              | 64.0%       |
|       | Imoto T, <i>et al.</i> 2010, JP <sup>19)</sup>     | 1,626           | 47.5       | Optimized cutoff  | Two or more of the following factors:<br>1) SBP $\geq$ 140 mmHg and/or DBP $\geq$ 90 mmHg<br>2) total cholesterol $\geq$ 220 mg/dL and/or HDL-cholesterol $<$ 40 mg/dL and/or triglyceride $\geq$ 150 mg/dL<br>3) fasting blood glucose $\geq$ 110 mg/dL                                                                    | 80.5          | 64.9%              | 69.3%       |

(Cont. Table 3)

| sex   | Authors, yr                                      | Participants, <i>n</i> | Age, years | WC threshold type | Criteria for cardiometabolic risk factor clustering                                                                                                                                                                                                                                                                                                            | WC cutoff, cm | Sensitivity                          | Specificity |
|-------|--------------------------------------------------|------------------------|------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------|-------------|
| Women | Oda E, <i>et al.</i> 2010 <sup>41)</sup>         | 1,442                  | 51.5       | Optimized cutoff  | NCEP-ATP III criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                    | 82            | 67.0%                                | 70.0%       |
|       | Ogawa D, <i>et al.</i> 2010 <sup>42)</sup>       | 2,161                  | 48.6       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 80            | 69.0%                                | 65%         |
|       | Ogawa D, <i>et al.</i> 2010 <sup>42)</sup>       | 2,161                  | 48.6       | Japanese criteria | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 90            | 29.3%                                | 91.5%       |
|       | Okauchi Y, <i>et al.</i> 2010 <sup>14)</sup>     | 548                    | 46.8       | Optimized cutoff  | Two or more of the following factors:<br>1) (SBP $\geq 130$ mmHg and/or DBP $\geq 85$ mmHg) or medical treatment<br>2) (fasting triglyceride $\geq 1.69$ mmol/L or postprandial $\geq 2.27$ mmol/L), and/or HDL-C $< 1.04$ mmol/L, or medical treatment<br>3) fasting serum glucose $\geq 6.10$ mmol/L or postprandial $\geq 7.77$ mmol/L or medical treatment | 81.5          | Maximal sensitivity plus specificity |             |
|       | Satoh H, <i>et al.</i> 2010 <sup>43)</sup>       | 1,434                  | 50         | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 81            | 75.0%                                | 70.0%       |
|       | Satoh H, <i>et al.</i> 2010 <sup>44)</sup>       | 1,452                  | 50         | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 83            | 71.0%                                | 75.0%       |
|       | Satoh H, <i>et al.</i> 2010 <sup>44)</sup>       | 1,452                  | 50         | Japanese criteria | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 90            | 47.0%                                | 88%         |
|       | Yamamoto Y, <i>et al.</i> 2010 <sup>45)</sup>    | 6,548                  | 51.4       | Optimized cutoff  | Three or more of the following factors:<br>1) SBP $\geq 130$ mmHg and/or DBP $\geq 85$ mmHg, or treatment<br>2) FPG $\geq 100$ mg/dL or HbA1c $\geq 5.2\%$ , or treatment<br>3) TG $\geq 150$ mg/dL and/or HDL-C $< 40$ mg/dL, or treatment                                                                                                                    | 79.1          | 70.6%                                | 67.6%       |
|       | Nakamura K, <i>et al.</i> 2011 <sup>47)</sup>    | 514                    | 52.9       | Optimized cutoff  | Two or more of the following factors:<br>1) SBP $\geq 130$ mmHg and/or DBP $\geq 85$ mmHg<br>2) HDL-cholesterol $< 40$ mg/dL and/or TG $\geq 150$ mg/dL<br>3) HbA1c (%) $\geq 5.5$                                                                                                                                                                             | 82            | 78.0%                                | 62.0%       |
|       | Nakamura K, <i>et al.</i> 2011 <sup>47)</sup>    | 514                    | 52.9       | Japanese criteria | Two or more of the following factors:<br>1) SBP $\geq 130$ mmHg and/or DBP $\geq 85$ mmHg<br>2) HDL-cholesterol $< 40$ mg/dL and/or TG $\geq 150$ mg/dL<br>3) HbA1c (%) $\geq 5.5$                                                                                                                                                                             | 90            | 38.0%                                | 88.0%       |
|       | Yamanaka C, <i>et al.</i> 2011 JP <sup>58)</sup> | 609                    | 40-74      | Japanese criteria | Two or more of the following factors:<br>1) SBP $\geq 130$ mmHg and/or DBP $\geq 85$ mmHg, or on treatment<br>2) TG $\geq 150$ mg/dL (fasting) or $\geq 300$ mg/dL (postprandial), and/or HDL-C $< 40$ mg/dL, or on treatment<br>3) Blood glucose $\geq 110$ mg/dL (fasting) or HbA1c $\geq 5.6\%$ (postprandial), or on treatment                             | 90            | 30.8%                                | 91.7%       |
|       | Matsushita Y, <i>et al.</i> 2012 <sup>48)</sup>  | 1,694                  | 55.8       | Optimized cutoff  | NCEP-ATP III criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                    | 84.1          | 66.4%                                | 63.2%       |
|       | Takahara M, <i>et al.</i> 2012 <sup>49)</sup>    | 961                    | 45.5       | Optimized cutoff  | Two or more of the following factors:<br>1) (SBP $\geq 130$ mmHg or DBP $\geq 85$ mmHg) or medical treatment<br>2) HDL-C $< 1.04$ mmol/L, (fasting triglyceride $\geq 1.69$ mmol/L or postprandial $\geq 2.27$ mmol/L), or medical treatment<br>3) fasting serum glucose $\geq 6.1$ mmol/L or postprandial $\geq 7.8$ mmol/L, or medical treatment             | 79            | Euclidean distance                   |             |
|       | Unno M, <i>et al.</i> 2012 <sup>50)</sup>        | 905                    | 52.6       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 84.5          | 68.6%                                | 66.8%       |
|       | Hosono A, <i>et al.</i> 2013 <sup>51)</sup>      | 1,138                  | 61.8       | Optimized cutoff  | NCEP-ATP III criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                    | 79            | 63%                                  | 57.6%       |
|       | Okazawa T, <i>et al.</i> 2013 <sup>52)</sup>     | 4,778                  | 48.2       | Optimized cutoff  | IDF criteria (excluding WC): risk factors $\geq 2$ in non-medicated participants                                                                                                                                                                                                                                                                               | 79.2          | 72.3%                                | 62.7%       |
|       | Tsukiyama H, <i>et al.</i> 2016 <sup>53)</sup>   | 166                    | 53.5       | Optimized cutoff  | Joint Interim Statement 2009 <sup>6)</sup> (excluding WC): risk factors $\geq 1$                                                                                                                                                                                                                                                                               | 78.0          | Maximal sensitivity plus specificity |             |
|       | Takase H, <i>et al.</i> 2020 <sup>54)</sup>      | 3,115                  | 37.9       | Optimized cutoff  | Japanese criteria (excluding WC): risk factors $\geq 2$                                                                                                                                                                                                                                                                                                        | 79.3          | 85.7%                                | 71.6%       |

\* Age: mean age of men and women combined.

Abbreviations: WC, waist circumference; NCEP-ATP III, National Cholesterol Education Program–Third Adult Treatment Panel; IDF, International Diabetes Federation; SBP, systolic blood pressure; DBP, diastolic blood pressure; HDL, high-density lipoprotein cholesterol; LDL, low-density lipoprotein cholesterol; FPG, fasting plasma glucose; TG, triglycerides.

**Table 4A.** Association of waist circumference cutoffs with cardiovascular disease risk in men

| Authors, yr                                                                                                                                            | Study population                                                              | Participants, <i>n</i> | Events, <i>n</i> | Age, years | Outcome                            | WC cutoff, cm      |                    | Adjusted variables                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------|------------------|------------|------------------------------------|--------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                        |                                                                               |                        |                  |            |                                    | 85cm<br>HR (95%CI) | 90cm<br>HR (95%CI) |                                                                                                                                                                         |
| Kokubo Y, <i>et al.</i> 2008 <sup>64)</sup><br>baseline: 1989-1994, mean<br>follow-up: 12.5 yrs)                                                       | Suita Study<br>(Urban residents)                                              | 2,492                  | 188              | 30-79y     | CVD                                | 0.97 (0.72-1.30)   | 1.18 (0.84-1.67)   | age                                                                                                                                                                     |
| Doi Y, <i>et al.</i> 2009 <sup>65)</sup><br>baseline: 1988, followed up for 14<br>yrs)                                                                 | Hisayama Study<br>(Suburban residents)                                        | 1,050                  | 131              | 58         | CVD                                | 1.22 (0.86-1.73)   | 1.81 (1.19-2.74)   | age                                                                                                                                                                     |
| Hata J, <i>et al.</i> 2010 <sup>66)</sup><br>baseline: 1988, followed up for 14<br>yrs)                                                                | Hisayama Study<br>(Suburban residents)                                        | 1,050                  | 66               | 58         | Ischemic stroke                    | 1.53 (0.94-2.50)   | 2.39 (1.38-4.14)   | age                                                                                                                                                                     |
| Okamura T, <i>et al.</i> 2011 <sup>67)</sup><br>baseline: 1989-1994, mean<br>follow-up: 13.0 yrs)                                                      | Suita Study<br>(Urban residents)                                              | 1,657                  | 29               | 30-64y     | Ischemic stroke                    | 1.46 (0.62-3.43)   |                    | age, smoking,<br>alcohol drinking<br>and LDL-C                                                                                                                          |
|                                                                                                                                                        |                                                                               |                        | 47               |            | CAD                                | 1.12 (0.55-2.25)   |                    |                                                                                                                                                                         |
|                                                                                                                                                        |                                                                               |                        | 654              | 65-79y     | Ischemic stroke                    | 1.00 (0.48-2.07)   |                    |                                                                                                                                                                         |
|                                                                                                                                                        |                                                                               |                        | 58               |            | CAD                                | 1.56 (0.85-2.87)   |                    |                                                                                                                                                                         |
| Yamazaki Y, <i>et al.</i> 2024 <sup>68)</sup><br>monitored for at least 3 years from<br>April 1, 2008, to July 31, 2016,<br>median follow-up: 5.2 yrs) | JMDC claims-based<br>database<br>(Insured person)                             | 330,051                | –                | 46         | CAD                                | 1.83 (1.68-2.00)   | 1.79 (1.63-1.96)   | age                                                                                                                                                                     |
|                                                                                                                                                        |                                                                               |                        | –                |            | Cerebrovascular<br>disease         | 1.41 (1.31-1.53)   | 1.52 (1.39-1.65)   |                                                                                                                                                                         |
|                                                                                                                                                        |                                                                               |                        | 3,934            |            | CAD/<br>cerebrovascular<br>disease | 1.59 (1.49-1.69)   | 1.66 (1.55-1.77)   |                                                                                                                                                                         |
| Hirokawa W, <i>et al.</i> 2010 <sup>69)</sup><br>baseline: 1996, mean follow-up:<br>10.0 yrs)                                                          | Employers of a metal<br>products factory in Toyama<br>Prefecture              | 2,685                  | 58               | 45.2       | CVD                                | 1.87 (1.07-3.26)   |                    | age, smoking,<br>alcohol drinking,<br>leisure-time<br>physical activity,<br>non-HDL-C, and<br>the three residual<br>factors (blood<br>pressure, lipids, and<br>glucose) |
| Sone H, <i>et al.</i> 2009 <sup>70)</sup><br>baseline: 1996, follow-up for 8<br>yrs)                                                                   | Japan Diabetes<br>Complications Study<br>(Diabetic patients; HbA1c<br>≥ 6.5%) | 771                    | 72               | 58.2       | CHD and/or<br>stroke               | 1.07 (0.65-1.74)   | 1.09 (0.62-1.93)   | age, smoking,<br>hypertension and<br>dyslipidemia                                                                                                                       |

Abbreviations: CVD, cardiovascular disease; CAD, coronary artery disease; LDL-C, low-density lipoprotein cholesterol; non-HDL-C, non-high-density lipoprotein cholesterol; CHD, coronary heart disease.

inconsistent findings (Table 5A)<sup>66, 72-75)</sup>. For example, in the Kyowa Study<sup>73)</sup>, ischemic stroke was associated with MetS according to the Japanese criteria (HR, 2.0 [95% CI, 1.1-3.6]), the IDF criteria for Asians (HR, 2.2 [95% CI, 1.2-3.9]), the modified NCEP-ATP III criteria 2 (HR, 2.0 [95% CI, 1.2-3.5]), and the estimated HRs were similar across definitions. In contrast, in the Hisayama Study<sup>66)</sup>, atherothrombotic and cardioembolic infarction were associated with MetS according to the IDF criteria for Asians (HR, 2.15 [95% CI, 1.06-4.34] and HR, 2.69 [95% CI, 1.27-5.68], respectively), whereas corresponding

associations were not observed according to the Japanese or modified NCEP-ATP III criteria, and the HR point estimates were lower.

In men, MetS defined according to the Japanese criteria (WC cutoff, 85 cm) was seldom associated with incident CVD, whereas definitions using a WC cutoff of 90 cm—including the modified Japanese, IDF criteria for Asians, and modified NCEP-ATP III criteria 2—more often showed associations with incident CVD and higher HR estimates (Table 5B)<sup>64-68, 76-79)</sup>. For example, in the Hisayama Study<sup>65)</sup>, the HR for incident CVD was 1.28 (95% CI, 0.86-1.91) when

**Table 4B.** Association of waist circumference cutoffs with cardiovascular disease risk in women

| Authors, yr                                                                                                                                           | Study population                                                              | Participants, <i>n</i> | Events, <i>n</i> | Age, years | Outcome                            | WC cutoff, cm      |                    | Adjusted variables                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------|------------------|------------|------------------------------------|--------------------|--------------------|-----------------------------------------------------------|
|                                                                                                                                                       |                                                                               |                        |                  |            |                                    | 80cm<br>HR (95%CI) | 90cm<br>HR (95%CI) |                                                           |
| Kokubo Y, <i>et al.</i> 2008 <sup>64</sup><br>baseline: 1989–1994, mean<br>follow-up: 12.5 yrs)                                                       | Suita Study<br>(Urban residents)                                              | 2,840                  | 129              | 30–79y     | CVD                                | 1.44 (1.00–2.07)   | 1.64 (1.09–2.46)   | age                                                       |
| Doi Y, <i>et al.</i> 2009 <sup>65</sup><br>baseline: 1988, followed up for 14<br>yrs)                                                                 | Hisayama Study<br>(Suburban residents)                                        | 1,402                  | 115              | 59         | CVD                                | 1.46 (0.99–2.16)   | 1.05 (0.68–1.62)   | age                                                       |
| Hata J, <i>et al.</i> 2010 <sup>66</sup><br>baseline: 1988, followed up for 14<br>yrs)                                                                | Hisayama Study<br>(Suburban residents)                                        | 1,402                  | 79               | 59         | Ischemic stroke                    | 1.16 (0.73–1.82)   | 1.13 (0.68–1.90)   | age                                                       |
| Okamura T, <i>et al.</i> 2011 <sup>67</sup><br>baseline: 1989–1994, mean<br>follow-up: 13.0 yrs)                                                      | Suita Study<br>(Urban residents)                                              | 2,057                  | 15               | 30–64y     | Ischemic stroke                    | 1.71 (0.59–4.94)   |                    | age, smoking,<br>alcohol drinking<br>and LDL-C            |
|                                                                                                                                                       |                                                                               |                        | 20               |            | CAD                                | 1.30 (0.52–3.20)   |                    |                                                           |
|                                                                                                                                                       |                                                                               |                        | 25               | 65–79y     | Ischemic stroke                    | 2.54 (1.01–6.40)   |                    |                                                           |
|                                                                                                                                                       |                                                                               |                        | 30               |            | CAD                                | 1.25 (0.59–2.62)   |                    |                                                           |
| Yamazaki Y, <i>et al.</i> 2024 <sup>68</sup><br>monitored for at least 3 years from<br>April 1, 2008, to July 31, 2016,<br>median follow-up: 5.2 yrs) | JMDC claims-based<br>database<br>(Insured person)                             | 235,028                | –                | 45         | CAD                                | 2.43 (1.70–3.47)   | 2.22 (1.50–3.28)   | age                                                       |
|                                                                                                                                                       |                                                                               |                        | –                |            | Cerebrovascular<br>disease         | 1.46 (1.27–1.67)   | 1.83 (1.54–2.17)   |                                                           |
|                                                                                                                                                       |                                                                               |                        | 893              |            | CAD/<br>cerebrovascular<br>disease | 1.61 (1.41–1.84)   | 1.93 (1.63–2.28)   |                                                           |
| Sone H, <i>et al.</i> 2009 <sup>70</sup><br>baseline: 1996, follow-up for 8<br>yrs)                                                                   | Japan Diabetes<br>Complications Study<br>(Diabetic patients; HbA1c<br>≥ 6.5%) | 653                    | 45               | 58.7       | CHD and/or<br>stroke               | 1.52 (0.81–2.85)   | 0.84 (0.29–2.38)   | age, smoking status,<br>hypertension, and<br>dyslipidemia |

Abbreviations: CVD, cardiovascular disease; CAD, coronary artery disease; LDL-C, low-density lipoprotein cholesterol; non-HDL-C, non-high-density lipoprotein cholesterol; CHD, coronary heart disease.

MetS was defined according to the Japanese criteria, 2.49 (95% CI, 1.57–3.94) according to the modified Japanese criteria, 1.96 (95% CI, 1.25–3.08) according to the IDF criteria for Asians, and 1.66 (95% CI, 1.14–2.43) according to the modified NCEP-ATP III criteria 2. Similar patterns were observed in the Suita Study<sup>64</sup> for CVD and stroke. In age-stratified analyses from the Suita Study<sup>64</sup>, the Japanese criteria were associated with increased CVD risk among men younger than 60 years (HR, 2.92 [95% CI, 1.54–5.55]), but not among those aged 60 years or older (HR, 1.06 [95% CI, 0.71–1.57]).

In women, MetS was associated with CVD risk in most studies, regardless of the diagnostic criteria applied (Table 5C)<sup>64–68, 77, 79</sup>. In the Suita Study<sup>64</sup>, both the Japanese criteria (HR, 6.25 [95% CI, 2.08–18.79]) and the modified NCEP-ATP III criteria 2 (HR, 2.96 [95% CI, 1.34–6.57]) were associated with increased CVD risk among women younger than 60

years. These findings suggest that the associations of MetS definitions incorporating WC with incident CVD may differ from those observed for WC alone, and that the clinical implications of clustered metabolic abnormalities may vary by age in ways not fully accounted for by the WC cutoff values.

Taken together, these findings suggest that the implications of WC in MetS-based CVD risk stratification extend beyond the appropriateness of the cutoff value itself. Associations of MetS definitions incorporating WC with incident CVD were not always parallel to those observed for WC alone and appeared to vary by sex and age. This pattern indicates that the clinical relevance of clustered metabolic abnormalities cannot be fully explained by WC thresholds alone. Therefore, requiring abdominal obesity as an essential component of MetS may not fully capture the CVD risk in some subgroups.

**Table 5A.** Association between metabolic syndrome criteria and risk of cardiovascular disease in the overall population

| Authors, yr                                | Study population                                                           | Participants, <i>n</i>              | Events, <i>n</i> | Age, years | Outcome                                             | metabolic syndrome criteria                           |                                    |                                    |                                    |                                    | Adjusted variables                                                                                                        |
|--------------------------------------------|----------------------------------------------------------------------------|-------------------------------------|------------------|------------|-----------------------------------------------------|-------------------------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|                                            |                                                                            |                                     |                  |            |                                                     | Japanese                                              | Modified Japanese                  | IDF for Asians                     | Modified NCEP 1                    | Modified NCEP 2 *                  |                                                                                                                           |
|                                            |                                                                            |                                     |                  |            |                                                     | WC; men 85, women 90<br>HR (95%CI)                    | WC; men 90, women 80<br>HR (95%CI) | WC; men 90, women 80<br>HR (95%CI) | WC; men 85, women 90<br>HR (95%CI) | WC; men 90, women 80<br>HR (95%CI) |                                                                                                                           |
| Saito I, <i>et al.</i> 2007 <sup>72)</sup> | Residents of O city, Ehime Prefecture (Rural residents)                    | 4,627                               | 88               | 40-74      | Stroke                                              | 0.82 (0.36-1.88)                                      |                                    |                                    |                                    |                                    | sex and age                                                                                                               |
| Chei CL, <i>et al.</i> 2008 <sup>73)</sup> | Residents of Kyowa area in Ibaraki Prefecture (Rural residents)            | 2,613                               | 42               | 40-69y     | Ischemic heart disease                              | 1.1 (0.5-2.2)                                         |                                    | 1.8 (0.9-3.4)                      |                                    | 2.1 (1.1-4.0)                      | age, time since last meal, cigarette smoking, alcohol intake and serum total cholesterol                                  |
|                                            |                                                                            |                                     | 73               |            | Total stroke                                        | 1.8 (1.1-3.1)                                         |                                    | 1.6 (1.0-2.7)                      |                                    | 1.7 (1.0-2.7)                      |                                                                                                                           |
|                                            |                                                                            |                                     | 54               |            | Ischemic stroke                                     | 2.0 (1.1-3.6)                                         |                                    | 2.2 (1.2-3.9)                      |                                    | 2.0 (1.2-3.5)                      |                                                                                                                           |
|                                            |                                                                            |                                     | 18               |            | Hemorrhagic stroke                                  | 1.4 (0.5-4.6)                                         |                                    | 0.7 (0.3-2.2)                      |                                    | 1.1 (0.4-2.8)                      |                                                                                                                           |
|                                            |                                                                            |                                     | 92               |            | Ischemic CVD                                        | 1.5 (1.0-2.4)                                         |                                    | 2.0 (1.3-3.2)                      |                                    | 2.0 (1.3-3.1)                      |                                                                                                                           |
| Hata J, <i>et al.</i> 2010 <sup>66)</sup>  | Hisayama Study (Suburban residents)                                        | 2,452 (Men: 1,050/<br>Women: 1,402) | 82               | ≥ 40       | Lacunar infarction                                  | 1.5 (0.82-2.72)                                       | 1.94 (1.13-3.32)                   | 1.65 (0.98-2.78)                   |                                    | 1.35 (0.83-2.22)                   | age, total cholesterol, proteinuria, electrocardiogram abnormalities, smoking habits, alcohol intake and regular exercise |
|                                            |                                                                            |                                     | 40               |            | Atherothrombotic infarction                         | 1.61 (0.76-3.43)                                      | 2.55 (1.25-5.18)                   | 2.15 (1.06-4.34)                   |                                    | 1.90 (0.99-3.63)                   |                                                                                                                           |
|                                            |                                                                            |                                     | 33               |            | Cardioembolic infarction                            | 1.96 (0.87-4.45)                                      | 3.94 (1.89-8.22)                   | 2.69 (1.27-5.68)                   |                                    | 2.20 (1.08-4.45)                   |                                                                                                                           |
| Saito I, <i>et al.</i> 2022 <sup>74)</sup> | Residents of O city, Ehime Prefecture (Rural residents)                    | 3,969                               | 376<br>99<br>268 | 40-74      | Stroke<br>Hemorrhagic stroke<br>cerebral infarction | 1.28(0.96-1.71)<br>0.89(0.44-1.77)<br>1.35(0.98-1.87) |                                    |                                    |                                    |                                    | sex and age                                                                                                               |
| He Y, <i>et al.</i> 2022 <sup>75)</sup>    | Participants of specific health checkups in Iwakura City, Aichi Prefecture | 6,496 (Men: 2,722/<br>Women: 3,774) | 149              | 62.5       | CVD                                                 | 1.6 (1.1-2.3)                                         |                                    |                                    |                                    |                                    | age, smoking status, alcohol habits, regular exercise, and LDL-C                                                          |

Abbreviations: IDF, International Diabetes Federation; NCEP, National Cholesterol Education Program Adult Treatment Panel III; HR, hazard ratio; CI, confidence interval; CVD, cardiovascular disease; LDL-C, low-density lipoprotein cholesterol.

\*For modified NCEP ATP III criteria 2, Chei *et al.* used fasting glucose ≥ 110 mg/dL, nonfasting glucose ≥ 140 mg/dL, or diabetes treatment and did not include treatment for lipid abnormalities. Hata *et al.* used fasting glucose ≥ 110 mg/dL and did not include treatment for lipid abnormalities.

**Table 5B.** Association between metabolic syndrome criteria and risk of cardiovascular disease in men

| Authors, yr                                   | Study population        | Participants, <i>n</i> | Events, <i>n</i> | Age, years | Outcome          | metabolic syndrome criteria   |                               |                               |                               |                               | Adjusted variables                 |
|-----------------------------------------------|-------------------------|------------------------|------------------|------------|------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|------------------------------------|
|                                               |                         |                        |                  |            |                  | Japanese                      | Modified Japanese             | IDF for Asians                | Modified NCEP 1 *             | Modified NCEP 2 **            |                                    |
|                                               |                         |                        |                  |            |                  | WC cutoff 85 cm<br>HR (95%CI) | WC cutoff 90 cm<br>HR (95%CI) | WC cutoff 90 cm<br>HR (95%CI) | WC cutoff 85 cm<br>HR (95%CI) | WC cutoff 90 cm<br>HR (95%CI) |                                    |
| Takeuchi H, <i>et al.</i> 2005 <sup>76)</sup> | Tanno and Sobetsu Study | 780                    | 49               | 60.3       | Cardiac Diseases |                               |                               |                               | 2.23 (1.14-4.34)              |                               | age, smoking and total cholesterol |

(Cont. Table 5B)

| Authors, yr                                                                                  | Study population                                    | Participants, <i>n</i> | Events, <i>n</i> | Age, years | Outcome                                    | metabolic syndrome criteria   |                               |                               |                               |                               | Adjusted variables                                                                                                                |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------|------------------|------------|--------------------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
|                                                                                              |                                                     |                        |                  |            |                                            | Japanese                      | Modified Japanese             | IDF for Asians                | Modified NCEP 1 *             | Modified NCEP 2 **            |                                                                                                                                   |
|                                                                                              |                                                     |                        |                  |            |                                            | WC cutoff 85 cm<br>HR (95%CI) | WC cutoff 90 cm<br>HR (95%CI) | WC cutoff 90 cm<br>HR (95%CI) | WC cutoff 85 cm<br>HR (95%CI) | WC cutoff 90 cm<br>HR (95%CI) |                                                                                                                                   |
| Kokubo Y, <i>et al.</i> 2008 <sup>64</sup><br>baseline: 1989-1994, mean follow-up: 12.5 yrs) | Suita Study (Urban residents)                       | 2,492                  | 188              | 30-79y     | Cardiovascular disease                     | 1.34 (0.96-1.87)              |                               |                               |                               | 1.75 (1.27-2.41)              | age, smoking and drinking status.                                                                                                 |
|                                                                                              |                                                     |                        | 42               | 30-59y     | Cardiovascular disease                     | 2.92 (1.54-5.55)              |                               |                               |                               | 1.94 (0.98-3.82)              |                                                                                                                                   |
|                                                                                              |                                                     |                        | 146              | 60-79y     | Cardiovascular disease                     | 1.06 (0.71-1.57)              |                               |                               |                               | 1.73 (1.20-2.48)              |                                                                                                                                   |
|                                                                                              |                                                     |                        | 78               | 30-79y     | Myocardial infarction                      | 1.51 (0.91-2.48)              |                               |                               |                               | 2.12 (1.31-3.43)              |                                                                                                                                   |
|                                                                                              |                                                     |                        | 110              | 30-79y     | All strokes                                | 1.27 (0.81-1.97)              |                               |                               |                               | 1.58 (1.02-2.43)              |                                                                                                                                   |
|                                                                                              |                                                     |                        |                  |            |                                            |                               |                               |                               |                               |                               |                                                                                                                                   |
| Ninomiya T, <i>et al.</i> 2007 <sup>77</sup><br>baseline: 1988, followed up for 14 yrs)      | Hisayama Study (Suburban residents)                 | 1,050                  | 158              | ≥ 40       | CVD                                        |                               |                               |                               |                               | 1.86 (1.32-2.62)              | age, proteinuria, electrocardiogram abnormalities, total cholesterol, smoking habits, alcohol intake, and regular exercise.       |
|                                                                                              |                                                     |                        | 78               |            | CHD                                        |                               |                               |                               |                               | 1.94 (1.19-3.17)              |                                                                                                                                   |
|                                                                                              |                                                     |                        | 94               |            | Stroke                                     |                               |                               |                               |                               | 1.92 (1.23-2.98)              |                                                                                                                                   |
|                                                                                              |                                                     |                        | 66               |            | Ischemic stroke                            |                               |                               |                               |                               | 1.68 (0.98-2.89)              |                                                                                                                                   |
|                                                                                              |                                                     |                        | 28               |            | Hemorrhagic stroke                         |                               |                               |                               |                               | 2.54 (1.18-5.49)              |                                                                                                                                   |
| Doi Y, <i>et al.</i> 2009 <sup>65</sup><br>baseline: 1988, followed up for 14 yrs)           | Hisayama Study (Suburban residents)                 | 1,050                  | 131              | 58         | CVD                                        | 1.28 (0.86-1.91)              | 2.49 (1.57-3.94)              | 1.96 (1.25-3.08)              |                               | 1.66 (1.14-2.43)              | age, serum total cholesterol, proteinuria, electrocardiogram abnormalities, alcohol intake, smoking habits, and regular exercise. |
| Sakurai M, <i>et al.</i> 2009 <sup>78</sup><br>baseline: 1996, followed up for 11 yrs)       | Employers of a products factory in Hokuriku region  | 2,903                  | 82               | 45.5       | CVD                                        | 2.26 (1.30-3.93)              |                               |                               |                               |                               | age, smoking habits, alcohol intake and regular exercise                                                                          |
| Hata J, <i>et al.</i> 2010 <sup>66</sup><br>baseline: 1988, followed up for 14 yrs)          | Hisayama Study (Suburban residents)                 | 1,050                  | 66               | 58         | Ischemic stroke                            | 1.32 (0.76-2.30)              | 3.07 (1.68-5.61)              | 2.66 (1.47-4.81)              |                               | 1.59 (0.93-2.74)              | age, total cholesterol, proteinuria, electrocardiogram abnormalities, smoking habits, alcohol intake and regular exercise         |
| Niwa Y, <i>et al.</i> 2010 <sup>79</sup><br>baseline: 1992-1995, mean follow-up: 11.2 yrs)   | Jichi Medical School Cohort Study (Rural residents) | 920                    | 54               | 56.5       | Stroke (excluding subarachnoid hemorrhage) | 1.89 (0.88-4.08)              |                               |                               |                               |                               | age, smoking status, and alcohol-drinking status                                                                                  |

(Cont. Table 5B)

| Authors, yr                                                                                                                                     | Study population                            | Participants, <i>n</i> | Events, <i>n</i> | Age, years | Outcome                      | metabolic syndrome criteria |                   |                  |                   |                    | Adjusted variables                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|------------------------|------------------|------------|------------------------------|-----------------------------|-------------------|------------------|-------------------|--------------------|------------------------------------------|
|                                                                                                                                                 |                                             |                        |                  |            |                              | Japanese                    | Modified Japanese | IDF for Asians   | Modified NCEP 1 * | Modified NCEP 2 ** |                                          |
|                                                                                                                                                 |                                             |                        |                  |            |                              | WC cutoff 85 cm             | WC cutoff 90 cm   | WC cutoff 90 cm  | WC cutoff 85 cm   | WC cutoff 90 cm    |                                          |
|                                                                                                                                                 |                                             |                        |                  |            |                              | HR (95%CI)                  | HR (95%CI)        | HR (95%CI)       | HR (95%CI)        | HR (95%CI)         |                                          |
| Okamura T, <i>et al.</i> 2011 <sup>67</sup><br>baseline: 1989-1994, mean follow-up: 13.0 yrs)                                                   | Suita Study (Urban residents)               | 1,657                  | 47               | 30-64y     | CAD                          |                             |                   |                  |                   | 1.21 (0.64-2.28)   | age, smoking, alcohol drinking and LDL-C |
|                                                                                                                                                 |                                             |                        | 29               |            | Ischemic stroke              |                             |                   |                  |                   | 3.24 (1.55-6.77)   |                                          |
|                                                                                                                                                 |                                             |                        | 58               | 65-79y     | CAD                          |                             |                   |                  |                   | 1.89 (1.11-3.21)   |                                          |
|                                                                                                                                                 |                                             |                        | 49               |            | Ischemic stroke              |                             |                   |                  |                   | 1.28 (0.67-1.36)   |                                          |
| Yamazaki Y, <i>et al.</i> 2024 <sup>68</sup><br>monitored for at least 3 years from April 1, 2008, to July 31, 2016, median follow-up: 5.2 yrs) | JMDC claims-based database (Insured person) | 330,051                | –                | 46         | CAD                          | 2.76 (2.52-3.02)            |                   | 2.28 (2.07-2.52) |                   | 2.81 (2.57-3.07)   | age                                      |
|                                                                                                                                                 |                                             |                        | –                |            | Cerebrovascular disease      | 1.90 (1.73-2.08)            |                   | 1.72 (1.56-1.89) |                   | 1.79 (1.64-1.95)   |                                          |
|                                                                                                                                                 |                                             |                        | 3,934            |            | CAD/ cerebrovascular disease | 2.25 (2.10-2.41)            |                   | 1.99 (1.85-2.14) |                   | 2.25 (2.10-2.40)   |                                          |
|                                                                                                                                                 |                                             |                        |                  |            |                              |                             |                   |                  |                   |                    |                                          |

Abbreviations: IDF, International Diabetes Federation; NCEP, National Cholesterol Education Program Adult Treatment Panel III; HR, hazard ratio; CI, confidence interval; CVD, cardiovascular disease; CHD, coronary heart disease; LDL-C, low-density lipoprotein cholesterol; CAD, coronary artery disease.

\* For modified NCEP ATP III criteria 1, Takeuchi *et al.* used fasting glucose > 110 mg/dL and excluded participants receiving treatment for hypertension, diabetes, or hyperlipidemia.

\*\* For modified NCEP ATP III criteria 2, Kokubo *et al.* used fasting glucose ≥ 110 mg/dL and did not include treatment for low HDL-C. Ninomiya *et al.* and Doi *et al.* used fasting glucose ≥ 110 mg/dL and did not include treatment for lipid abnormalities. Okamura *et al.* did not include treatment for lipid abnormalities. Yamazaki *et al.* did not include treatment for diabetes.

**Table 5C.** Association between metabolic syndrome criteria and risk of cardiovascular disease in women

| Authors, yr                                                                                  | Study population                    | Participants, <i>n</i> | Events, <i>n</i> | Age, years | Outcome                | metabolic syndrome criteria |                   |                 |                 |                   | Adjusted variables                                                                                                          |
|----------------------------------------------------------------------------------------------|-------------------------------------|------------------------|------------------|------------|------------------------|-----------------------------|-------------------|-----------------|-----------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
|                                                                                              |                                     |                        |                  |            |                        | Japanese                    | Modified Japanese | IDF for Asians  | Modified NCEP 1 | Modified NCEP 2 * |                                                                                                                             |
|                                                                                              |                                     |                        |                  |            |                        | WC cutoff 90 cm             | WC cutoff 80 cm   | WC cutoff 80 cm | WC cutoff 90 cm | WC cutoff 80 cm   |                                                                                                                             |
|                                                                                              |                                     |                        |                  |            |                        | HR (95%CI)                  | HR (95%CI)        | HR (95%CI)      | HR (95%CI)      | HR (95%CI)        |                                                                                                                             |
| Ninomiya T, <i>et al.</i> 2007 <sup>77</sup><br>baseline: 1988, followed up for 14 yrs)      | Hisayama Study (Suburban residents) | 1,402                  | 149              | ≥ 40       | CVD                    |                             |                   |                 |                 | 1.70 (1.22-2.36)  | age, proteinuria, electrocardiogram abnormalities, total cholesterol, smoking habits, alcohol intake, and regular exercise. |
|                                                                                              |                                     |                        | 47               |            | CHD                    |                             |                   |                 |                 | 2.86 (1.56-5.24)  |                                                                                                                             |
|                                                                                              |                                     |                        | 115              |            | Stroke                 |                             |                   |                 |                 | 1.50 (1.03-2.19)  |                                                                                                                             |
|                                                                                              |                                     |                        | 79               |            | Ischemic stroke        |                             |                   |                 |                 | 1.78 (1.13-2.79)  |                                                                                                                             |
|                                                                                              |                                     |                        | 36               |            | Hemorrhagic stroke     |                             |                   |                 |                 | 0.99 (0.48-2.05)  |                                                                                                                             |
| Kokubo Y, <i>et al.</i> 2008 <sup>64</sup><br>baseline: 1989-1994, mean follow-up: 12.5 yrs) | Suita Study (Urban residents)       | 2,840                  | 129              | 30-79y     | Cardiovascular disease | 2.20 (1.31-3.68)            |                   |                 |                 | 1.90 (1.31-2.77)  | age, smoking and drinking status.                                                                                           |
|                                                                                              |                                     |                        | 29               | 30-59y     | Cardiovascular disease | 6.25 (2.08-18.79)           |                   |                 |                 | 2.96 (1.34-6.57)  |                                                                                                                             |
|                                                                                              |                                     |                        | 100              | 60-79y     | Cardiovascular disease | 1.80 (1.01-3.20)            |                   |                 |                 | 1.70 (1.12-2.59)  |                                                                                                                             |
|                                                                                              |                                     |                        | 39               | 30-79y     | Myocardial infarction  | 2.70 (1.15-6.35)            |                   |                 |                 | 2.77 (1.44-5.32)  |                                                                                                                             |
|                                                                                              |                                     |                        | 90               | 30-79y     | All strokes            | 2.05 (1.07-3.92)            |                   |                 |                 | 1.62 (1.02-2.58)  |                                                                                                                             |
|                                                                                              |                                     |                        |                  |            |                        |                             |                   |                 |                 |                   |                                                                                                                             |

(Cont. Table 5C)

| Authors, yr                                                                                                                                      | Study population                                    | Participants, <i>n</i> | Events, <i>n</i> | Age, years | Outcome                      | metabolic syndrome criteria |                   |                  |                 |                  | Adjusted variables                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------|------------------|------------|------------------------------|-----------------------------|-------------------|------------------|-----------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                  |                                                     |                        |                  |            |                              | Japanese                    | Modified Japanese | IDF for Asians   | Modified NCEP 1 | Modified NCEP 2* |                                                                                                                                   |
|                                                                                                                                                  |                                                     |                        |                  |            |                              | WC cutoff 90 cm             | WC cutoff 80 cm   | WC cutoff 80 cm  | WC cutoff 90 cm | WC cutoff 80 cm  |                                                                                                                                   |
|                                                                                                                                                  |                                                     |                        |                  |            |                              | HR (95%CI)                  | HR (95%CI)        | HR (95%CI)       | HR (95%CI)      | HR (95%CI)       |                                                                                                                                   |
| Doi Y, <i>et al.</i> 2009 <sup>65)</sup><br>baseline: 1988, followed up for 14 yrs)                                                              | Hisayama Study (Suburban residents)                 | 1,402                  | 115              | 59         | CVD                          | 1.89 (1.15–3.10)            | 2.27 (1.55–3.32)  | 1.79 (1.23–2.60) |                 | 1.88 (1.30–2.74) | age, serum total cholesterol, proteinuria, electrocardiogram abnormalities, alcohol intake, smoking habits, and regular exercise. |
| Hata J, <i>et al.</i> 2010 <sup>66)</sup><br>baseline: 1988, followed up for 14 yrs)                                                             | Hisayama Study (Suburban residents)                 | 1,402                  | 79               | 59         | Ischemic stroke              | 2.09 (1.17–3.75)            | 2.21 (1.39–3.51)  | 1.74 (1.11–2.73) |                 | 1.73 (1.10–2.71) | age, total cholesterol, proteinuria, electrocardiogram abnormalities, smoking habits, alcohol intake and regular exercise         |
| Niwa Y, <i>et al.</i> 2010 <sup>79)</sup><br>baseline: 1992–1995, mean follow-up: 11.2 yrs)                                                      | Jichi Medical School Cohort Study (Rural residents) | 1,285                  | 42               | 56.0       | Stroke                       | 7.24 (2.82–18.58)           |                   |                  |                 |                  | age, smoking status, and alcohol-drinking status                                                                                  |
| Okamura T, <i>et al.</i> 2011 <sup>67)</sup><br>baseline: 1989–1994, mean follow-up: 13.0 yrs)                                                   | Suita Study (Urban residents)                       | 2,057                  | 20               | 30–64y     | CAD                          |                             |                   |                  |                 | 4.44 (1.73–11.4) | age, smoking, alcohol drinking and LDL-C                                                                                          |
|                                                                                                                                                  |                                                     |                        | 15               |            | Ischemic stroke              |                             |                   |                  |                 | 3.99 (1.34–11.8) |                                                                                                                                   |
|                                                                                                                                                  |                                                     |                        | 30               | 65–79y     | CAD                          |                             |                   |                  |                 | 1.98 (0.95–4.11) |                                                                                                                                   |
|                                                                                                                                                  |                                                     |                        | 25               |            | Ischemic stroke              |                             |                   |                  |                 | 2.03 (0.92–4.48) |                                                                                                                                   |
| Yamazaki Y, <i>et al.</i> 2024 <sup>68)</sup><br>monitored for at least 3 years from April 1, 2008, to July 31, 2016, median follow-up: 5.2 yrs) | JMDC claims-based database (Insured person)         | 235,028                | –                | 45         | CAD                          | 4.21 (2.68–6.63)            |                   | 3.84 (2.68–5.51) |                 | 3.98 (2.77–5.72) | age                                                                                                                               |
|                                                                                                                                                  |                                                     |                        | –                |            | Cerebrovascular disease      | 2.22 (1.72–2.87)            |                   | 1.98 (1.67–2.35) |                 | 2.02 (1.71–2.38) |                                                                                                                                   |
|                                                                                                                                                  |                                                     |                        | 893              |            | CAD/ cerebrovascular disease | 2.66 (2.11–3.37)            |                   | 2.32 (1.97–2.72) |                 | 2.30 (1.97–2.70) |                                                                                                                                   |
|                                                                                                                                                  |                                                     |                        |                  |            |                              |                             |                   |                  |                 |                  |                                                                                                                                   |

Abbreviations: IDF, International Diabetes Federation; NCEP, National Cholesterol Education Program Adult Treatment Panel III; HR, hazard ratio; CI, confidence interval; CVD, cardiovascular disease; CHD, coronary heart disease; LDL-C, low-density lipoprotein cholesterol; CAD, coronary artery disease.

\* For modified NCEP ATP III criteria 2, Kokubo *et al.* used fasting glucose  $\geq 110$  mg/dL and did not include treatment for low HDL-C. Ninomiya *et al.* and Doi *et al.* used fasting glucose  $\geq 110$  mg/dL and did not include treatment for lipid abnormalities. Okamura *et al.* did not include treatment for lipid abnormalities. Yamazaki *et al.* did not include treatment for diabetes.

### Development of Waist Circumference Criteria in International MetS Definitions

The origin of WC criteria in international definitions of MetS can be traced to the NCEP ATP III criteria<sup>4)</sup>. In this definition, the WC cutoff values were set at 102 cm for men and 88 cm for women, corresponding to a BMI of approximately 30 kg/m<sup>2</sup> (obesity)<sup>80)</sup>. These thresholds were derived from data

obtained in European populations. The IDF definition introduced ethnicity-specific WC cutoff values while retaining abdominal obesity as a mandatory component<sup>5)</sup>. In this framework, cutoff values of 90 cm for men and 80 cm for women were proposed for Asian populations. This proposal was partly informed by the statement from the World Health Organization (WHO) Expert Consultation that Asian populations tend to develop diabetes and

cardiovascular risk factors at lower BMI levels than European populations, and that WC should be used to refine action levels based on BMI<sup>81</sup>). Accordingly, the IDF adopted these thresholds through expert consensus based on heterogeneous data sources rather than from a single definitive epidemiological study. Therefore, the WC cutoff values for Asian populations in the IDF definition were established pragmatically, reflecting the limited availability of robust epidemiological evidence. In the harmonized criteria proposed in 2009, abdominal obesity was no longer treated as a mandatory component, but it was instead considered one of the five diagnostic components of metabolic syndrome<sup>6</sup>). At the same time, the statement recommended the use of population- and country-specific waist circumference cut-off values.

### Challenges in CVD Risk Stratification Based on the Waist Circumference Criteria in Japan

The requirement for abdominal obesity in definitions, such as those of the IDF and the Japanese criteria, reflects the shared concept that central obesity, as an indicator of visceral fat accumulation, is closely linked to insulin resistance and strongly associated with the clustering of other cardiometabolic risk factors underlying metabolic syndrome. This relationship is largely explained by the metabolic effects of visceral adiposity. Visceral adiposity has been shown to contribute to metabolic abnormalities through several mechanisms, including the increased portal release of free fatty acids, leading to hepatic insulin resistance; enhanced secretion of pro-inflammatory cytokines, such as tumor necrosis factor- $\alpha$  and interleukin-6; and reduced secretion of adiponectin, which decreases insulin sensitivity<sup>1</sup>).

In Japan, the Specific Health Checkup program implements CVD high-risk screening based on the Japanese diagnostic criteria for metabolic syndrome<sup>8</sup>). Under this system, individuals are classified as high-risk and targeted for health guidance (lifestyle intervention) when obesity is present together with additional cardiometabolic risk factors. In this context, recent studies have demonstrated that clustering of cardiometabolic risk factors is associated with increased cardiovascular risk, irrespective of obesity status<sup>82, 83</sup>). Furthermore, the WC cutoff value for women (90 cm) in the Japanese criteria is relatively high compared with international standards and has been considered controversial<sup>65, 68</sup>). The findings from the present literature review also suggest that, for women, lower WC thresholds than the current criterion may improve the prediction of clustering of cardiometabolic risk factors.

In the context of the Japanese health check-up system, which applies the Japanese MetS criteria for CVD high-risk screening, two potential issues have been raised. First, the current waist circumference cut-off values, particularly the 90 cm threshold for women, may be too high to effectively identify high-risk individuals. Second, accumulating evidence suggests that obesity may not be a prerequisite for elevated CVD risk, as individuals without obesity, but with multiple cardiometabolic risk factors can have higher cardiovascular risk than those without such risk factors.

In summary, these findings highlight the potential limitations of CVD risk stratification strategies that rely heavily on the WC criteria in the Japanese population. Although central obesity remains an important marker of cardiometabolic risk, accumulating evidence suggests that substantial cardiovascular risk may also be present among individuals without abdominal obesity, but with multiple metabolic abnormalities. In addition, the currently recommended WC threshold for women may not optimally identify individuals at elevated risk. These observations suggest that future refinement of CVD risk screening strategies in Japan may benefit from approaches that place greater emphasis on the overall burden and clustering of cardiometabolic risk factors in addition to abdominal obesity. Further epidemiological studies are warranted to determine screening strategies that more accurately identify individuals at increased cardiovascular risk in the Japanese population.

### Conflicts of Interest

The authors have no conflicts of interest to declare.

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## Supplemental Methods

### 1. Search Strategies

#### 1-1. Search Strategy 1 for Theme 1: Development and Evaluation of WC Criteria Based on Visceral Fat Area

| No | Search query                                                                                                                                                                                                                                                                                                       | Results |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| #1 | ("intra abdominal fat"[MeSH Terms] OR "Abdominal Fat"[Mesh:NoExp] OR "visceral fat"[Title/Abstract] OR "Abdominal Fat"[TIAB])                                                                                                                                                                                      | 24,424  |
| #2 | ("metabolic syndrome"[MeSH Terms] OR "metabolic syndrome"[Title/Abstract] OR ("waist circumference"[MeSH Terms] OR "waist circumference"[Title/Abstract]) OR ("visceral obesity"[Title/Abstract] OR "abdominal obesity"[Title/Abstract] OR "central obesity"[Title/Abstract] OR "obesity, abdominal"[MeSH Terms])) | 124,077 |
| #3 | #1 AND #2                                                                                                                                                                                                                                                                                                          | 7,594   |
| #4 | "Japan"[MeSH Terms] OR "japan*"[Title/Abstract]                                                                                                                                                                                                                                                                    | 349,378 |
| #5 | #3 AND #4                                                                                                                                                                                                                                                                                                          | 441     |

#### 1-2. Search Strategy 2 for Theme 2: Predictability of WC for Clustering of Cardiometabolic Risk Factors

| No  | Search query                                                                                                                                       | Results   |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| #1  | "metabolic syndrome"[MeSH Terms] OR "metabolic syndrome"[Title/Abstract]                                                                           | 81,908    |
| #2  | "waist circumference"[MeSH Terms] OR "waist circumference"[Title/Abstract]                                                                         | 43,947    |
| #3  | "visceral obesity"[Title/Abstract] OR "abdominal obesity"[Title/Abstract] OR "central obesity"[Title/Abstract] OR "obesity, abdominal"[MeSH Terms] | 21,081    |
| #4  | #1 OR #2 OR #3                                                                                                                                     | 124,030   |
| #5  | "sensitivity and specificity"[MeSH Terms] OR "sensitiv*"[Title/Abstract]                                                                           | 2,357,263 |
| #6  | "cutoff*"[Title/Abstract] OR "cut off*"[Title/Abstract]                                                                                            | 189,106   |
| #7  | "reproducibility of results"[MeSH Terms] OR "reference values"[MeSH Terms] OR "risk factors"[MeSH Terms]                                           | 1,717,563 |
| #8  | #5 OR #6 OR #7                                                                                                                                     | 3,849,360 |
| #9  | #4 AND #8                                                                                                                                          | 46,538    |
| #10 | "Japan"[MeSH Terms] OR "japan*"[Title/Abstract]                                                                                                    | 349,295   |
| #11 | #9 AND #10                                                                                                                                         | 1,249     |

#### 1-3. Search Strategy 3 for Themes 3 and 4: Association Between WC Cutoffs and CVD Risk, and Association Between MetS Criteria and CVD Risk

| No  | Search query                                                                                                                                                                                                                                                                                                 | Results   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| #1  | "metabolic syndrome"[MeSH Terms] OR "metabolic syndrome"[Title/Abstract] OR "waist circumference"[MeSH Terms] OR "waist circumference"[Title/Abstract] OR "visceral obesity"[Title/Abstract] OR "abdominal obesity"[Title/Abstract] OR "central obesity"[Title/Abstract] OR "obesity, abdominal"[MeSH Terms] | 124,052   |
| #2  | "cardiovascular diseases"[MeSH Terms] OR "coronary artery disease*"[Title/Abstract] OR "coronary heart disease*"[Title/Abstract] OR "coronary disease*"[Title/Abstract]                                                                                                                                      | 2,968,088 |
| #3  | "cerebrovascular disorders"[MeSH Terms] OR "stroke"[Title/Abstract] OR "cerebrovascular"[Title/Abstract]                                                                                                                                                                                                     | 686,912   |
| #4  | brain Infarction[mh] OR myocardial Infarction[mh]                                                                                                                                                                                                                                                            | 249,824   |
| #5  | infarct*[tiab] AND (myocard*[tiab] OR cerebral[tiab] OR brain[tiab] OR lacunar[tiab] OR atherothrombotic[tiab])                                                                                                                                                                                              | 327,006   |
| #6  | "ischemi*"[Title/Abstract] AND ("myocard*"[Title/Abstract] OR "coronary"[Title/Abstract] OR "cardio*"[Title/Abstract] OR "cerebral"[Title/Abstract] OR "brain"[Title/Abstract])                                                                                                                              | 254,486   |
| #7  | #2 OR #3 OR #4 OR #5 OR #6                                                                                                                                                                                                                                                                                   | 3,190,855 |
| #8  | #1 AND #7                                                                                                                                                                                                                                                                                                    | 33,821    |
| #9  | "Japan"[MeSH Terms] OR japan*[Title/Abstract]                                                                                                                                                                                                                                                                | 349,332   |
| #10 | #8 AND #9                                                                                                                                                                                                                                                                                                    | 853       |

### 2. Eligibility Criteria

#### 2-1. Eligibility Criteria for Theme 1

- General population or individuals with type 2 diabetes
- Japanese adults (aged  $\geq 18$  years)
- At least 50 participants in each reported sex-specific group
- Waist circumference corresponding to visceral fat area
- Reported threshold values for both visceral fat area and waist circumference

#### 2-2. Eligibility Criteria for Theme 2

- General population or individuals with type 2 diabetes
- Japanese adults (aged  $\geq 18$  years)
- At least 50 participants in each reported sex-specific group

- Waist circumference evaluated for identifying clustering of at least two cardiometabolic risk factors
- Cardiometabolic risk factors defined as hypertension or elevated blood pressure, diabetes or hyperglycemia, and dyslipidemia including both triglycerides and HDL cholesterol
- Reported waist circumference cutoffs derived from ROC analysis

### *2-3. Eligibility Criteria for Theme 3*

- General population or individuals with type 2 diabetes
- Japanese adults (aged  $\geq 18$  years)
- Waist circumference as the exposure of interest
- Reported hazard ratios with 95% confidence intervals
- Incident cardiovascular disease events (studies

using CVD mortality alone as the outcome were excluded)

- Reported number of incident events

### *2-4. Eligibility Criteria for Theme 4*

- General population or individuals with type 2 diabetes
- Japanese adults (aged  $\geq 18$  years)
- Metabolic syndrome as the exposure of interest, defined using waist circumference (excluding definitions using body mass index as a substitute)
- Reported hazard ratios with 95% confidence intervals
- Incident cardiovascular disease events (studies using CVD mortality alone as the outcome were excluded)
- Reported number of incident events